

Chapter 3 — Hardware

3.1 Computer Architecture

3.1.1 The CPU

CPU (Central Processing Unit) / microprocessor — responsible for execution/processing of all instructions & data in a computer application. Often installed as an **integrated circuit** on a single microchip.

CPU consists of: **Control Unit (CU)** + **Arithmetic & Logic Unit (ALU)** + **Registers & Buses**

3.1.2 Von Neumann Architecture

Developed by John von Neumann (mid-1940s) — the "stored program computer".

Main features:

- Concept of a CPU (processor)
- CPU can access memory directly
- Memories store **both** programs and data
- Stored programs = instructions executed in **sequential order**

Von Neumann Diagram (simplified)

[Address bus] ↑↓ MAR ↔ PC ↔ CU ↔ ALU ↔ ACC
[Data bus] ↑↓ MDR ↔ CIR
[Control bus] ↔ CU
All inside → **CPU**

CPU Components

ALU — carries out arithmetic (+, −, shifting) & logic (AND, OR) ops. Multiplication/division done via sequence of addition, subtraction, shifts. A computer can have >1 ALU.

Control Unit (CU) — reads instruction from memory (address in PC), interprets via Fetch-Decode-Execute cycle, sends signals along control bus, synchronises data flow.

System clock — produces timing signals on control bus for synchronisation — without it, computer would crash.

RAM = Immediate Access Store (IAS) — CPU moves data/programs from backing store into RAM temporarily since RAM read/write is much faster.

Registers (Table 3.1)

Register	Abbr.	Function
Current Instruction Register	CIR	stores current instruction being decoded/executed
Accumulator	ACC	stores data temporarily during ALU calculations
Memory Address Register	MAR	stores address of memory location being read/written
Memory Data/Buffer Register	MDR	stores data just read from / about to be written to memory
Program Counter	PC	stores address of next instruction to be fetched

Exam-Style Practice

1. Name three registers found in the CPU and describe the function of each. [6]
2. Explain the role of the Control Unit in the Fetch-Decode-Execute cycle. [3]

System Buses & Memory Read/Write

CPU ↔ Memory ↔ Input/Output ↔ Ports
via → Control bus / Address bus / Data bus (= System bus)

READ operation (example)

Want to read location **1111 0001**:

1. Address written into **MAR**: 1111 0001
2. 'Read signal' sent to memory
3. Contents put into **MDR**: 0101 1011

WRITE operation (example)

Want to write **1001 0101** into location **1111 1101**:

1. Data written into **MDR**: 1001 0101
2. Address written into **MAR**: 1111 1101
3. 'Write signal' sent → value stored at that location

(System) Buses

Address bus — carries addresses; **unidirectional** (CPU→memory only). Width matters: 16-bit bus → $2^{16}=65\,536$ locations; 32-bit → 4 294 967 296 locations.

Data bus — **bidirectional**; carries address/instruction/numerical value. Wider bus = larger **word length** transported = better performance.

Control bus — **bidirectional**; carries signals from CU to all components; usually 8-bit wide.

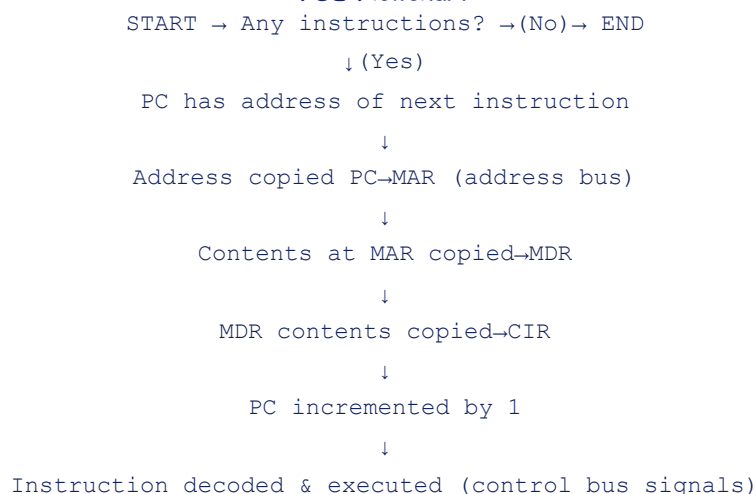
3.1.2 Fetch-Decode-Execute Cycle

Fetch — next instruction fetched from address in MAR → stored in MDR → copied to CIR → PC incremented by 1.

Decode — instruction interpreted so it can be processed.

Execute — CPU sends decoded instruction as control signals to appropriate components — carried out in logical sequence.

FDE Flowchart



More instructions? →(Yes, loop back) / (No→END)

Exam-Style Practice

1. Describe the steps that take place during the fetch stage of the FDE cycle. [4]
2. State the purpose of the address bus and explain why it is unidirectional. [3]
3. Complete: bus width, _____, _____ affect overall computer performance. [3]

3.1.3 Cores, Cache & Internal Clock

Factors affecting CPU performance:

1. **Bus width** (address + data) — wider = faster
2. **Clock speed** — typical value 3.5GHz = 3.5 billion cycles/sec. **Overclocking** (via BIOS) risky: unsynchronised operation / crashes / overheating.
3. **Cache memory** — located *within* CPU (faster than RAM); stores frequently used instructions/data; CPU checks cache before RAM. Bigger cache = better performance.
4. **Number of cores** — dual-core (1 channel), quad-core (6 channels). More cores ≠ proportionally more performance (communication overhead between cores reduces the gain).

Higher clock speed alone does NOT guarantee better overall performance!

3.1.4 Instruction Set

Instruction = **opcode** (what operation) + **operand** (data acted on / register).

Instruction set = limited number of opcodes CPU understands. Example: **X86** instruction set (Intel Pentium, AMD Athlon share nearly identical X86 sets despite different electronic designs).

Do NOT confuse instruction sets with programming code — program code needs interpreters/compilers to convert to instruction set. Examples of ops: ADD, JMP, LDA.

3.1.5 Embedded Systems

Embedded system = hardware + software combo designed for a specific set of functions.

Type	Description
Microcontroller	CPU + RAM + ROM + peripherals all on ONE chip
Microprocessor	ONLY CPU on chip — no RAM/ROM/peripherals
System on Chip (SoC)	usually CPU + memory + I/O ports + secondary storage on ONE microchip

Note: a computer is **NOT** an embedded system (multi-functional, varies software).

Programmable devices upgraded via: (1) connect to computer & download update, (2) automatic update via Wi-Fi/satellite/cellular.

Benefits & Drawbacks (Table 3.3)

Benefits	Drawbacks
small size, easy to fit into devices	hard to upgrade for new tech
low cost to manufacture	troubleshooting = specialist task
dedicated to one task → simple interface, often no OS needed	interface can appear simple but actually confusing (e.g. cooker clock)
consume very little power	vulnerable to hackers/viruses if internet-connected
can be controlled remotely	devices often thrown away rather than repaired
fast real-time reaction, feedback oriented	contributes to "throw-away society"

Examples: Motor vehicles, Set-top box, Security, Lighting, Vending, Washing machines

Vending machine microcontroller inputs: keypad, tilt sensors, position sensors, coin counter, temperature sensors. Outputs: actuators (dispense/chill pump), display, wireless modem (sales data back to operator).

Exam-Style Practice

1. Distinguish between a microcontroller and a microprocessor. [2]
2. Give two benefits and two drawbacks of using embedded systems in devices. [4]
3. A security system uses an embedded system. Describe the inputs and outputs involved. [6]

3.2 Input & Output Devices

3.2.1 Barcode Scanners

Barcode = series of dark/light parallel lines of varying thickness; different codes for left-side and right-side digits (allows scan in any direction).

Scanning process:

1. Red laser/LED reads barcode
2. Light reflected back — dark areas reflect little/no light
3. Reflected light read by photoelectric cell sensors
4. Pattern generated → converted to digital data

Then: barcode number = **key field** → stock record looked up → price sent to POS → stock count reduced by 1 → compared to re-order level → auto-reorder if needed.

Barcode Advantages

To Management	To Customers
faster price changes	faster checkout
up-to-date sales info	fewer charging errors
no need to price every item	itemised bill
automatic stock control	cost savings passed on
tracks buying habits (loyalty cards)	fresher food (sell-by tracking)

QR Codes

Matrix of dark squares on light background; holds up to **4296 characters** (vs 30 digits max for barcode).

3 large corner squares = alignment; small corner square = ensures correct size/angle of camera shot.

Advantages vs Barcode	Disadvantages vs Barcode
hold much more info	more than one QR format exists
built-in error checking → fewer errors	can transmit malicious code ("attagging")
read by phone camera (no laser needed)	
easy to transmit as text/image	
can be encrypted	

Frame QR codes — have a "canvas area" for logos/graphics.

Exam-Style Practice

1. Describe how data is read from a barcode when it is scanned. [4]
2. Give two advantages of QR codes over traditional barcodes. [2]

Digital Cameras

Controlled by embedded system: adjusts shutter speed, focus, flash, aperture, image size, red-eye removal.

Charge Couple Device (CCD) — light-sensitive cell of millions of sensors (photodiodes) that convert light→electricity. Each sensor = **pixel**.

Analogue to Digital Converter (ADC) — converts electric charges into brightness levels (e.g. 8-bit ADC = $2^8=256$ levels).

24-bit RGB: each pixel has R (0-255), G (0-255), B (0-255) values.

lens → CCD → ADC → digital image (binary)
(timing device controls shutter speed etc.)

Keyboards

Each character has an **ASCII value**. Membrane/circuit board at base of keys — key press completes a circuit → CPU identifies key → refers to index file for character.

Drawback: **Repetitive Strain Injury (RSI)** — ergonomic keyboards help.

Microphones

Sound waves → diaphragm vibrates → coil (around cone) moves through magnetic field of permanent magnet → induces electric current (analogue) → sound card ADC → digital signal for storage.

Optical Mouse

Red LED light bounced off surface → CMOS picks up reflection → generates electric pulses → DSP calculates coordinates → cursor moved.

Optical vs Mechanical	Wired vs Bluetooth mouse
no moving parts → more reliable	wired: no signal loss
dirt can't get trapped	wired: cheaper (no batteries)
works on any surface	wired: fewer environmental issues

Exam-Style Practice

1. Explain how a digital camera converts an image into digital data. [4]
2. Describe how an optical mouse detects movement. [4]

2D and 3D Scanners

2D scanner steps: cover raised → document on glass → cover closed → bright light illuminates document → scan head moves, mirrors direct image to lens → lens focuses onto **CCD** (light→electric current) → software produces digital image.

OCR (Optical Character Recognition) — converts scanned text into editable text file format.

3D scanners — scan solid objects using x,y,z coordinates → digital 3D image; used in CAD / 3D printing.
Technologies: lasers, magnetic resonance, white light.

Application — CT Scanners: build 3D image via thin 2D "slices" (tomography). CT=X-rays, MRI=radio frequencies, SPECT=gamma rays.

Touch Screens

Type	How it works	Key Adv.	Key Disadv.
Capacitive	finger disturbs electrostatic field on conductive layer; microcontroller calculates coordinates	clear image, durable, projective type = multi-touch	surface type only works with bare finger/stylus; sensitive to EM radiation
Infrared	grid of IR beams; finger breaks beam(s); sensors detect reduction	multi-touch, durable even if scratched/cracked	sensitive to water/moisture; light interference
Resistive	2 resistive layers separated by air; touch = contact = circuit completed	good vs dust/water; works with gloves/stylus	low sensitivity, no multi-touch, poor in sunlight, scratch-prone

Exam-Style Practice

1. Explain how a capacitive touch screen detects the position of a finger touch. [4]
2. Compare infrared and resistive touch screen technologies. [4]

3.2.2 Output Devices

Actuators

Mechanical/electromechanical device (relay, solenoid, motor) that converts electrical signal into physical motion, e.g. solenoid — current in coil creates magnetic field → moves plunger → operates valve/switch.

Light Projectors

DLP (Digital Light Projector)	LCD Projector
uses DMD chip — millions of micro mirrors (ON/OFF tilt)	white light → dichromic mirrors split R/G/B → 3 LCD screens → recombined via prism
higher contrast, quieter, smaller, better in dusty rooms	sharper image, better colour saturation, more energy efficient
images can "shadow" on motion; no grey component	lower contrast; limited longevity; screens yellow over time

Inkjet vs Laser Printers

Inkjet	Laser
print head sprays ink droplets (thermal bubble OR piezoelectric)	uses dry powder toner + static electricity; prints whole page at once
cheap, good for one-off photos/small colour jobs	fast, high volume, large toner + paper trays

Laser printing steps: drum given +ve charge → laser removes +ve charge in pattern (leaves -ve areas = image) → +ve toner sticks to -ve areas → -ve paper rolled over drum → toner transfers to paper → paper charge removed → **fuser** (heated rollers) melts toner onto paper → discharge lamp resets drum.

3D Printers

Additive manufacturing (build layer by layer) vs **Subtractive** (remove material, e.g. CNC).

- **Direct 3D printing** — inkjet tech, head moves L-R and up-down
- **Binder 3D printing** — 2 passes: dry powder then binder (glue) sprayed

Process: CAD design → prepared in 3D-printer format → printer set up → object built layer by layer (~0.1mm) → object removed & prepared/cured.

Exam-Style Practice

1. Describe the stages involved when a laser printer prints a document. [6]
2. State one difference between direct 3D printing and binder 3D printing. [2]

LED, LCD & OLED Screens

LED screen	LCD screen	OLED screen
tiny LEDs (R/G/B); used for bright outdoor displays	liquid crystal pixels; needs backlighting (LED or old CCFL)	organic material between cathode & anode; self-lit, no backlight needed

CAUTION: TVs sold as "LED TV" are usually LCD screens backlit by LEDs — not pure LED screens!

OLED advantages: thinner/lighter/flexible, brighter, no backlighting (less power), can bend/fold, ~170° field of view.

(Loud)Speakers

digital data → DAC (digital→analogue) → amplifier → loudspeaker → sound

Electric current → coil around iron core becomes temporary electromagnet → attracted/repelled by permanent magnet as current varies → cone vibrates → sound waves produced.

3.2.3 Sensors

Sensors = input devices that read/measure physical (analogue) properties. Need **ADC** to convert to digital for computer use. **DAC** used when computer controls analogue devices (motor/valve).

Feedback = output of microprocessor affects the next input from sensor.

Sensor	Example Application
Temperature	central heating control, greenhouse
Moisture	soil monitoring in greenhouse
Humidity	building/factory monitoring
Light	street lights, car headlights
Infrared (active)	car wipers (rain), security (beam break)
Infrared (passive)	security (body heat), freezer temp
Pressure	lorry weighing, nuclear reactor gas pressure
Acoustic/sound	footstep detection, pipe leak detection
Gas	pollution/O ₂ /CO ₂ monitoring
pH	soil acidity, chemical process
Magnetic field	ABS braking systems
Accelerometer	airbag deployment, phone orientation
Proximity	phone screen off near ear
Flow rate	hospital inhalers, gas pipes
Level	petrol tank, tablet powder levels

Exam-Style Practice

1. Explain the difference between an ADC and a DAC, giving one use of each. [4]

2. Name a suitable sensor for monitoring soil acidity in a greenhouse and explain how it would be used.
[3]

Monitoring vs Control Systems

Monitoring	Control
computer just "watches" — warning/alarm if out of range	computer sends signals to valves/motors/actuators to change the system
no effect on the process	output affects next input → feedback loop

Sensor → ADC → Microprocessor compares vs stored values
→ MONITOR: alarm/screen msg if out of range
→ CONTROL: signal to valve/motor (via DAC) if out of range → feedback loop

Key Worked Examples

Street lighting: light sensor → ADC → microprocessor samples every minute → if reading < stored value → lamp ON (stays 30 min before re-sampling to avoid flicker); if $\geq$ stored value → lamp OFF.

ABS brakes: magnetic field sensor detects wheel rotating too slowly → microprocessor compares to other 3 wheels → reduces braking pressure to that wheel → checked several times/second — felt as "judder".

Central heating: temp sensor → ADC → compared to pre-set value → if temp < pre-set → signal via DAC → open gas valve + turn on water pump.

Chemical process: temp & pH sensors → if temp < 70°C heater ON; if pH > 3.5 add acid.

Exam-Style Practice

1. Describe, using a named example, the difference between a monitoring system and a control system. [6]
2. A greenhouse uses a humidity sensor to control window motors. Describe how this control system works. [5]

3.3 Data Storage

Memory & Storage

→ Primary memory: RAM, ROM

→ Secondary (internal): HDD, SSD

→ Secondary (external): DVD/CD, Blu-ray, USB stick, removable HDD

Primary memory	Secondary storage
directly addressable by CPU	NOT directly addressable by CPU
RAM, ROM, cache	all non-volatile; HDD/SSD/DVD/stick/Blu-ray; internal or external

3.3.1 RAM

RAM — **volatile** (loses data when powered off); can be read/written; stores data/files/apps/OS currently in use.

DRAM	SRAM
transistors + capacitors	uses flip-flops
needs constant refresh (every ~15 microseconds)	no refresh needed
cheaper, less power, higher capacity	faster (25ns access vs DRAM 60ns)
used as main memory	used in CPU cache

ROM

ROM — **non-volatile**, permanent, read-only. Stores BIOS/start-up instructions (bootstrap).

RAM	ROM
temporary	permanent
volatile	non-volatile
read + write	read only
stores current data/programs/OS in use	stores BIOS/start-up data

Example (remote toy car): ROM = factory settings, start-up routines, button mappings. RAM = user-programmed routines, incoming remote-control data.

Exam-Style Practice

1. State four differences between RAM and ROM. [4]
2. Explain why DRAM needs to be refreshed but SRAM does not. [3]

3.3.3 Magnetic, Optical & Solid-State Storage

HDD (Hard Disk Drive): magnetic platters spin (~7000 rps); read-write heads (electromagnets); data stored in **sectors & tracks**. **Latency** = time for a data block to rotate to the head. Fragmentation over time → slower access → fixed by defragmentation.

SSD (Solid State Drive): no moving parts; NAND/NOR chips; data via floating-gate & control-gate transistors (CMOS); non-volatile.

SSD benefits over HDD
more reliable (no moving parts), lighter, no "spin up" time, lower power, runs cooler, thinner, faster data access

Drawback: SSD endurance — limited write cycles (~20GB writes/day over 3 yrs typical rating)

Optical Media

Disk	Laser	Wavelength	Construction
CD	Red	780nm	single 1.2mm layer
DVD (dual-layer)	Red	650nm	two 0.6mm layers
Blu-ray (single)	Blue	405nm	single 1.2mm layer
Blu-ray (dual)	Blue	405nm	two 0.6mm layers

Shorter wavelength = smaller pits/lands = **more storage**. Blu-ray transfer rate 36Mbps vs DVD 10Mbps. Blu-ray has built-in encryption (anti-piracy).

Capacity: DVD dual-layer 4.7GB · Blu-ray single 27GB · Blu-ray dual 50GB.

Exam-Style Practice

1. Explain why an SSD accesses data faster than a HDD. [3]
2. Explain why Blu-ray discs can store more data than DVDs. [3]

3.3.4 Virtual Memory

Virtual memory = **RAM + swap space on HDD/SSD**. Used when RAM runs out (prevents system crash). Data moved in/out using **paging** (fixed-length blocks).

Oldest RAM data mapped out to HDD/SSD to free space for new program — managed via a "map" that tracks program↔RAM address mapping.

Benefits	Drawback
programs can be larger than physical RAM	Disk thrashing — excessive swapping slows/halts processing (thrash point); more RAM/SSD reduces this
reduces need to buy more RAM	
no wasted memory on unused data	

3.3.5 Cloud Storage

Public cloud	Private cloud	Hybrid cloud
client & provider = different companies	behind company firewall; client & provider = same entity	mix of both

Data redundancy — same data stored on multiple servers (in case of repair/maintenance).

Benefits	Drawbacks
access anywhere, anytime (needs internet)	slow/unstable internet = access problems
no need to carry storage device	can be costly for large capacity/data transfer
remote backup / disaster recovery	provider could fail → risk of data loss
almost unlimited capacity	data security concerns (relinquish control)

Exam-Style Practice

1. Explain what is meant by virtual memory and why it is needed. [4]
2. Give two benefits and two drawbacks of using cloud storage. [4]

3.4 Network Hardware

3.4.1 NIC

Network Interface Card (NIC) — allows device to connect to a network; contains **MAC address** (set at manufacture). **WNIC** = wireless version (antenna, microwaves).

3.4.2 MAC Address

48 bits — 6 groups of hex digits: **NN-NN-NN-DD-DD-DD** (manufacturer code + device serial number). Changes if NIC is replaced.

UAA	LAA
Universally Administered — set by manufacturer (most common)	Locally Administered — manually changed (e.g. to bypass MAC filters, match mainframe formats)

3.4.3 IP Address

IPv4	IPv6
32-bit, 4 groups of denary digits e.g. 254.25.28.77	128-bit, 8 groups of hex digits e.g. A8FB:7A88:...

IPv6 advantages: no address collision risk, built-in authentication, more efficient routing.

MAC address	IP address
physical device address on network	global address on internet
unique per device, set by manufacturer	may not be unique; assigned by ISP (DHCP)
48 bits, UAA or LAA	32-bit (IPv4) or 128-bit (IPv6); static or dynamic

Dynamic IP	Static IP
assigned by DHCP each time device connects; changes each time; more private	permanently assigned; used for servers/FTP/databases; fully traceable; more expensive to maintain

Exam-Style Practice

1. Describe the structure of a MAC address. [3]
2. Give two differences between a static IP address and a dynamic IP address. [2]

3.4.4 Routers

Router — connects/routes data packets between different networks (e.g. LAN↔WAN); converts data between protocols/formats of different networks.

Computer/Server ↔ SWITCH ↔ ROUTER ↔ Internet / LAN or WAN

Broadband routers sit behind a **firewall** (protects LAN computers). Router inspects packet → uses IP address to find correct switch → uses **MAC destination address** to deliver to correct device. If MAC not found on that switch → passed to another switch on same network. Routers can be wired or wireless.

Quick Recap — Whole Chapter

Topic	One-line memory hook
Von Neumann	CPU + direct memory access + stored programs + sequential execution
FDE Cycle	Fetch (MAR→MDR→CIR, PC+1) → Decode → Execute
Cache vs RAM	cache = inside CPU, faster, smaller
RAM vs ROM	RAM=volatile/read-write; ROM=non-volatile/read-only
HDD vs SSD	HDD=magnetic/moving parts; SSD=NAND chips/no moving parts
Virtual memory	RAM + swap space on disk (via paging)
MAC vs IP	MAC=physical/manufacturer; IP=logical/ISP assigned

Exam-Style Practice

1. Explain how a router uses IP and MAC addresses to deliver a data packet to the correct device. [4]
2. Explain the role of a firewall in relation to a broadband router. [2]